

Tableau — Introductory Notes

1. Introduction to Tableau

Tableau is a **data visualization and business intelligence (BI) tool** used to connect to data, analyze it, and present the results through interactive charts, dashboards, and reports.

Tableau helps users:

- Turn raw data into meaningful visualizations.
- Identify trends, patterns, and relationships.
- Create interactive dashboards.
- Share insights with other users.
- Perform analysis without needing extensive programming knowledge.

Example: A company can use Tableau to visualize monthly sales, compare regional performance, and identify its best-selling products.

2. Tableau vs Power BI

Feature	Tableau	Power BI
Developer	Salesforce	Microsoft
Main strength	Data visualization and interactive analysis	BI, reporting and Microsoft ecosystem integration
Ease of visualization	Very flexible and highly visual	Easy to use, especially for Microsoft users
Data modeling	Good	Strong, especially with DAX
Integration	Connects to many data sources	Excellent integration with Excel, Azure, Microsoft 365
Calculations	Calculated fields, table calculations	DAX
Dashboards	Highly interactive	Highly interactive
Typical users	Analysts, data professionals, business users	Business analysts, organizations using Microsoft tools
Deployment	Tableau Cloud, Tableau Server, Desktop	Power BI Service, Desktop, Report Server
Cost	Generally higher depending on edition/licensing	Often considered cost-effective, with Microsoft licensing options

In short: Tableau is particularly known for **visual analytics and flexible dashboards**, while Power BI is strongly integrated with the **Microsoft ecosystem and data-modeling tools**.

3. Where Tableau is Used

Tableau is used across many industries:

- **Banking & Finance:** financial analysis, risk analysis, investment reporting
- **Healthcare:** patient and hospital analytics
- **Retail:** sales, inventory, customer behavior
- **Education:** student performance and institutional reporting
- **Manufacturing:** production, quality, and supply-chain analysis
- **Marketing:** campaign performance and customer analytics
- **Government:** public-sector data analysis
- **Human Resources:** employee and workforce analytics

Common applications include:

- Sales dashboards
- Financial reports
- KPI monitoring
- Customer analysis
- Market research
- Operational reporting
- Performance tracking

4. Versions / Products of Tableau

Tableau is available in several products and deployment options:

1. Tableau Desktop

Used to connect to data, perform analysis, and create worksheets, dashboards, and stories.

2. Tableau Server

An organization-hosted platform for publishing, managing, and sharing Tableau content.

3. Tableau Cloud

A cloud-hosted version used to publish, share, and collaborate on Tableau content without managing your own Tableau Server infrastructure.

4. Tableau Public

A free platform for creating and publicly sharing visualizations. Data published here is generally accessible publicly.

5. Tableau Reader

A tool historically used for viewing Tableau workbooks. Its role has become less central as Tableau's cloud/server sharing options have evolved.

5. Tableau Connectors

A **connector** allows Tableau to connect to a data source.

Common connectors

File-based sources

- Microsoft Excel
- CSV
- Text files
- JSON
- PDF

Databases

- MySQL
- Microsoft SQL Server
- PostgreSQL
- Oracle
- Snowflake

Cloud/data platforms

- Google BigQuery
- Amazon Redshift
- Microsoft Azure data services
- Salesforce

Tableau can work with both **live connections** and **extracts**.

- **Live connection:** Tableau queries the original data source when analysis is performed.
- **Extract:** Tableau stores a copy/optimized subset of the data for analysis.

6. Tableau Interface

The main parts of the Tableau interface include:

- **Data Pane:** Displays tables, dimensions, and measures.
- **Columns Shelf:** Used to determine what appears along the horizontal direction.
- **Rows Shelf:** Used to determine what appears along the vertical direction.
- **Marks Card:** Controls properties such as color, size, label, shape, and detail.
- **Filters Shelf:** Used to filter data.
- **Pages Shelf:** Allows analysis across different values sequentially.
- **Show Me:** Provides suggested visualization types.
- **Worksheet:** Area where individual visualizations are created.
- **Dashboard:** Combines multiple worksheets into one interactive view.
- **Story:** Combines worksheets and dashboards to present a sequence of insights.

7. Simple Steps to Create a Visualization

For example, suppose we have an Excel file containing **Sales, Product, Region, and Date**.

Step 1 — Connect to data

Open Tableau → select **Connect to Data** → choose Excel/CSV/database.

Step 2 — Select a worksheet

After connecting, open a new worksheet.

Step 3 — Choose fields

Drag a **dimension** such as **Region** to the **Rows** shelf.

Step 4 — Add a measure

Drag **Sales** to the **Columns** shelf.

Step 5 — Choose a chart

Use **Show Me** or the Marks card to select a suitable chart, such as a bar chart.

Step 6 — Add formatting

Drag fields to **Color**, **Label**, or **Size** to make the visualization more informative.

Step 7 — Apply filters

Drag a field such as **Year** or **Product** to the **Filters** shelf.

Step 8 — Create a dashboard

Click **New Dashboard** and drag your worksheets onto it.

Step 9 — Save or publish

Save the workbook or publish it to Tableau Server/Cloud/Public as appropriate.

Simple example

Data → **Connect** → **Worksheet** → **Select fields** → **Choose chart** → **Filter** → **Format** → **Dashboard** → **Share**

This is the basic workflow for creating a visualization in Tableau.